Estimating GDP from Emissions Data:
Do Pollution and Economic Growth Go Together?

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Statistics and Machine Learning: Final Project Report
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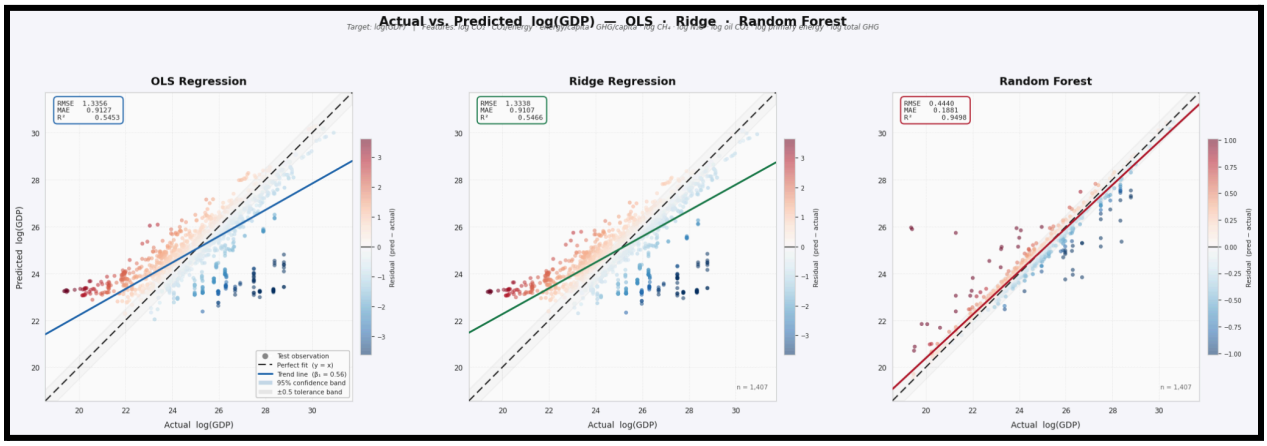


With our current economic infrastructure, emissions are heavily tied to GDP data, and countries with higher emission will have high GDPs because it represents productivity in the economy. It is important to understand emissions' role not just in our economy but also in our environment. This graph shows three regressions run on emissions data to predict GDP. The strongest regression, the Random Forest model demonstrates the strong positive relationship between emissions and GDP. This can also be seen through a correlation matrix of the data. We must invest in scale back our emissions and reliance on polluting practices as further damage to our environment will have larger downside effects than the current gain from ignoring pollution warnings.

Problem Statement

In this study we will be looking at how emissions and GDP are tied together and explore whether GDP can be predicted from emissions data with the hope to understand how changing our emissions could affect GDP given our current infrastructure. Environmental economists and policy makers are the primary stakeholders. This research can allow them to understand the quantitative relationship of emissions and GDP. With this they will be able to critically analyze which air pollutants provide the strongest economic benefit, and properly consider the social and public cost of emissions to the economic gain. Balancing social, public, and private interests in an

ongoing, controversial issue that gaining a measurable insight in would help inform policies.

Data Description

The data is from Our World in Data, a project built on a collaboration between researchers at the University of Oxford and the non-profit organisation Global Change Data Lab. This dataset is the premiere source for country emissions, used by the World Bank and many others for its comprehensive and current data. It contains emission, population, and GDP data since 1750 for all major countries, territories, and regions.

Before processing the dataset had 50411 observations (rows) and 79 features (columns). Over half the data was missing, with inconsistent records for many smaller nations and older years. Many of the data was MNAR (Missing Not at Random). Most of the missing values were countries or regions that have poor records and low means to collect records. Also, much of the missing data could be older data that is uncollectable since the dataset starts in 1750, but most emission data was not measured until 1990. Using Claude, I cleaned the dataset and interpolated much of the missing data. The cleaned dataset had 7035 observations and 18 features. Using a Tukey Fences test, I found 12196 total outliers, but decided to keep them. Any outliers would have likely been high-income and high-emission countries as GDP and emissions are heavily right skewed. These outliers are important to our model as they represent the extreme emission countries who this would be most relevant to.

The correlation matrix showed strong correlations between many features, especially with features and GDP. This shows a strong relationship and supports our initial claim. With all the features either strongly correlated or near-neutral, it gives an interesting feature set to our regression. Features were selected through manual column parsing. Redundant columns were dropped such as ‘total_ghg_excluding_lucf’ vs. ‘total_ghg’. Finally any positive and non-normalization column was log-transformed given their skew.
The processed data had the features: 'co2_log', 'co2_per_unit_energy', 'energy_per_capita', 'ghg_per_capita', 'methane_log', 'nitrous_oxide_log', 'oil_co2_log', 'primary_energy_consumption_log', and 'total_ghg_log', regressed on 'gdp_log'.

Methodology

Four models were run in: a baseline OLS regression, a Ridge regression to see the effects of the multicollinearity, a Random Forest regression as the advanced reasoning model that can understand the complex relationship between features, and a Gradient Boosting regression as comparison to the Random Forest.

All models were evaluated with an 80/20 train-test split. Data given to the OLS and Ridge models were scaled to mitigate the effects of large values on the coefficients. Data was cross validated with RMSE, MAE, and R^2. These metrics allow us to tell the predictive capabilities and variance of the models.

The OLS regression tells us the linear relationship and gives us a baseline for other models. An OLS model retains all features it is given, without striking coefficients, giving a clearer relationship between variables. It works well when the p/n ratio is lower and there is low multicollinearity. Looking at the correlation matrix, there is high multicollinearity which made me inclined to try a Ridge regression. Ridge regressions are another type of linear regression where coefficients of features are shrunk to prevent overfitting. Features are adjusted by the hyperparameter alpha, which controls the regularization, bias, and variance. Reducing the effects of correlated variables creates a more stable and reliable regression. The Random Forest regression that is trained on randomly sampled data put through decision trees. This model is able to reduce overfitting by averaging predictors from different trees, understand complex non-linear relationships without scaling the features, and it is able rank which variables contributed to the model the most.

Results

Model	RMSE	MAE	R^2
OLS	1.3356	0.9127	0.5453
Ridge	1.3338	0.9107	0.5466
Random Forest	0.4440	0.1881	0.9498
Gradient Boosting	0.7646	0.4791	0.8510

Based on our cross-validation results (R^2 = 0.9568 +/- 0.0046), we estimate that the Random Forest model is able to correctly predict 95.68% of the variance in log-GDP, giving it the strongest predictive power out of our model while have the lowest errors. Ridge did not perform as well as expected, having marginally better cross-validation metrics than its OLS counterpart.

Feature importance analysis showed that 'primary_energy_consumption_log' accounted for over 50% of the model's predictive power, with the rest of the features contributing about 10% or less. This concentration shows the strong collinearity in the data. To ensure that the feature was truly a strong predictor, I introduced the Gradient Boosting model. Gradient Boosting may cycle between strongly correlated predictors, and running the regression we saw little change in the feature importance. Given that the features were strongly correlated to the target variables, it is likely that the model is learning that pattern of the most correlated variables and using the remaining features to tune up the remaining residuals.

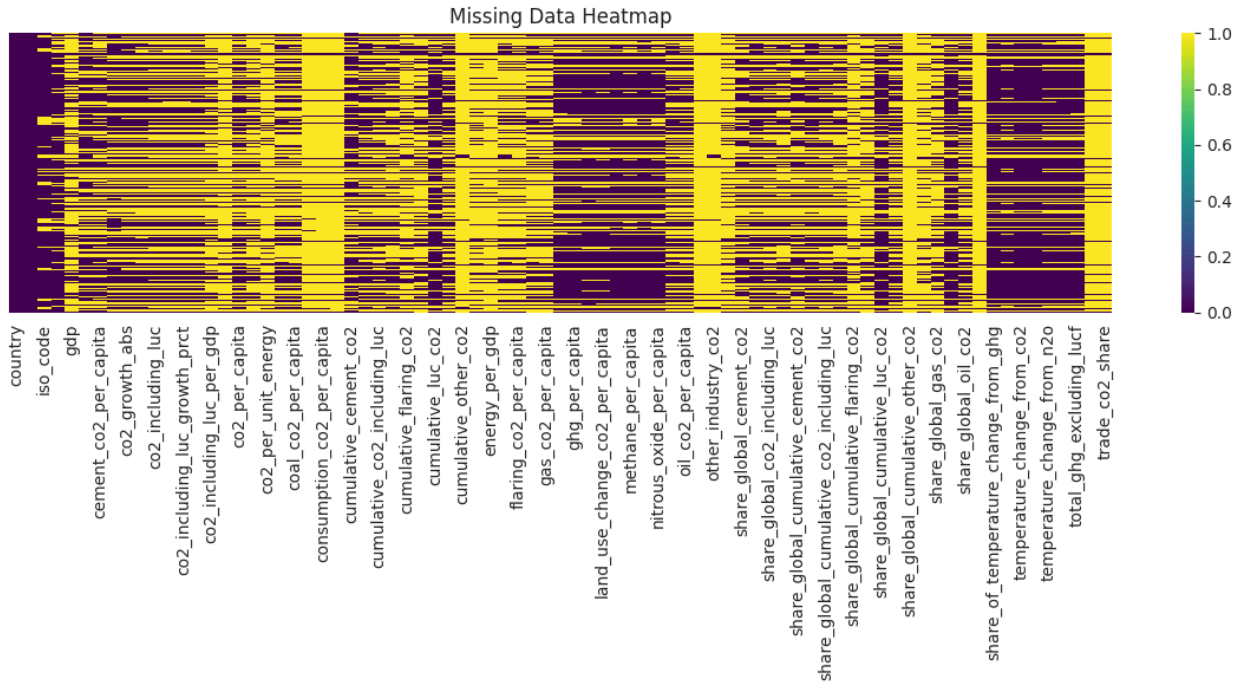
Recommendation

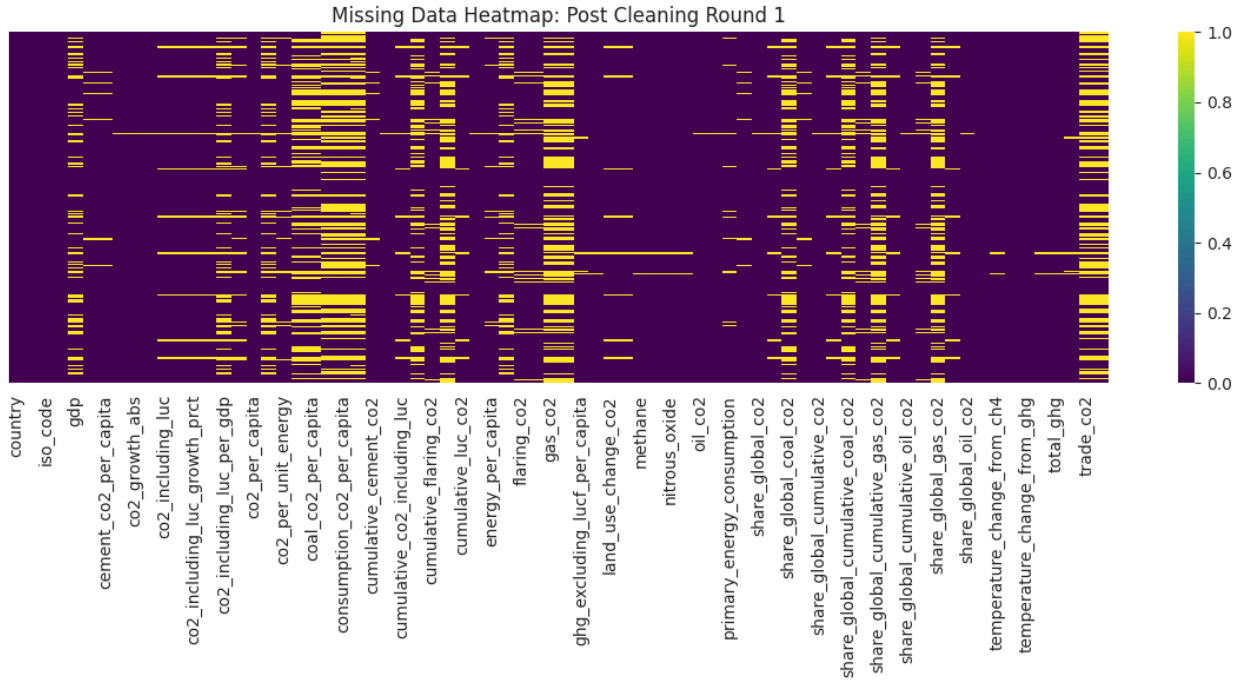
Based on our cross-validation results (R^2 = 0.9568 +/- 0.0046), we estimate that the Random Forest model is able to correctly predict 95.68% of the variance in log-GDP, giving it strong predictive power. The primary limitation is the dominance and multicollinearity of the features. We recommend policy makers use the model for predictive applications on the effects of lower emissions on GDP with the caveat that policy should be examined through fixed-effects strategies that are able to isolate the effects.

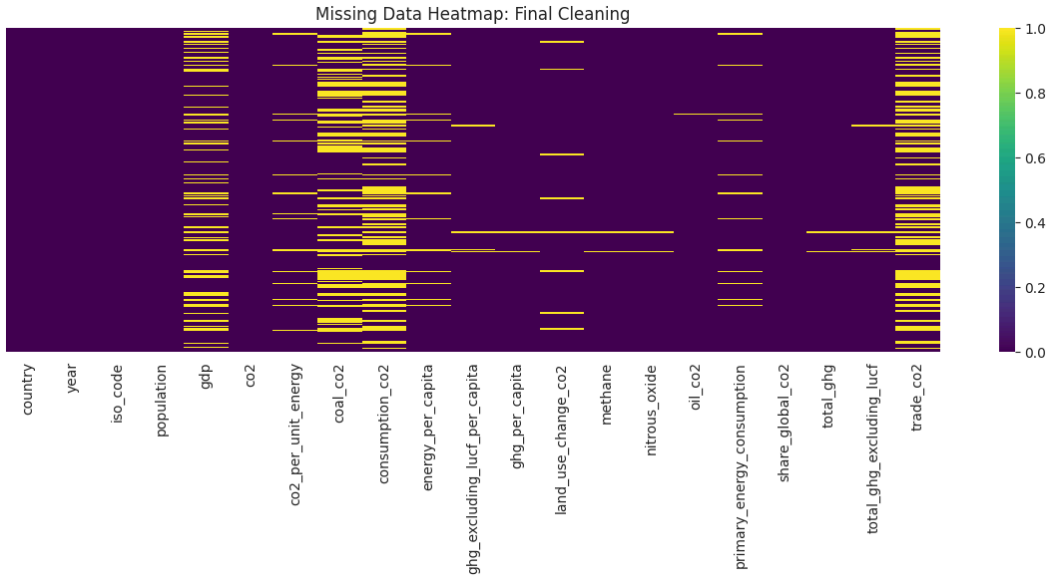
Deciding the costs and benefits of emissions is a difficult task. On one hand, emissions cause permanent damage to the environment, having large consequences including global warming/climate change, ecological damage, polluted air, and more. We have already seen many of these consequences take a toll on the earth. On the other hand, emissions are a huge, and sometimes unavoidable part of economic progress. We have relied on fossil fuels, polluting industries, and consumerist ideals to run homes, businesses, and the global progress. Not being able to emit any pollutants would halt the global economy. While some polluting parties may be able to find alternative methods, there cannot operate without pollutants. Global shipping, electricity, commodities, financial markets, AI, and more would collapse if they were not able to pollute at all. This research project revealed that our current economy is intertwined with emissions. Global emissions could not be separated out from GDP, and their collinearity depicts a detrimental symbiotic relationship.

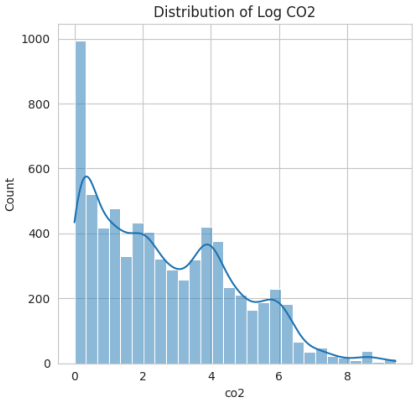
Environmental scientists estimate that we are mere years away from polluting the environment beyond repair. You must think of pollution as an economic bubble. The data, both the models and the correlation structure of the features show that emissions are tied closely to GDP. This is especially true for stronger economies where the high GDP and high emissions go hand-in-hand. Our current regulations do not consider these risks. With the recent repeal of the 2009 Greenhouse Gas Endangerment Finding, which removed the legal basis for regulating greenhouse gases, we can see that long-term costs are not being considered. This inaction of change incurs a multi-trillion dollar global cost, far outweighing the cost of adjusting infrastructure in the short term.

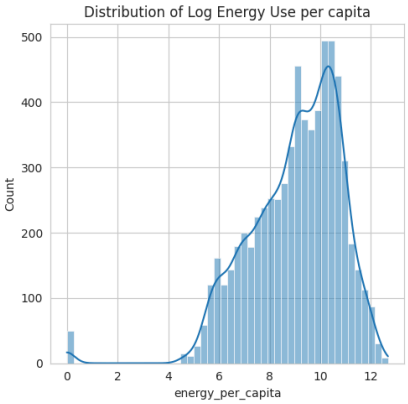
Graph Appendix

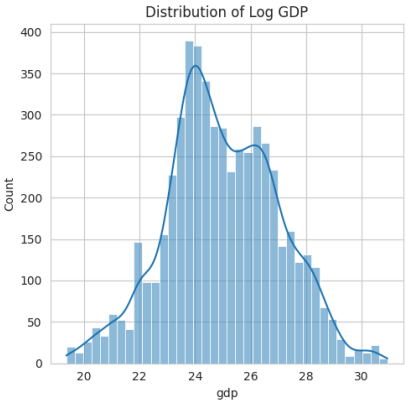


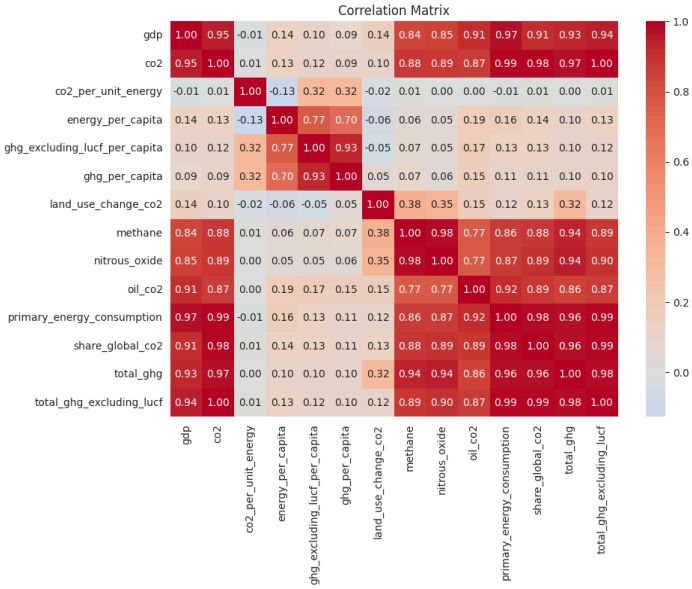


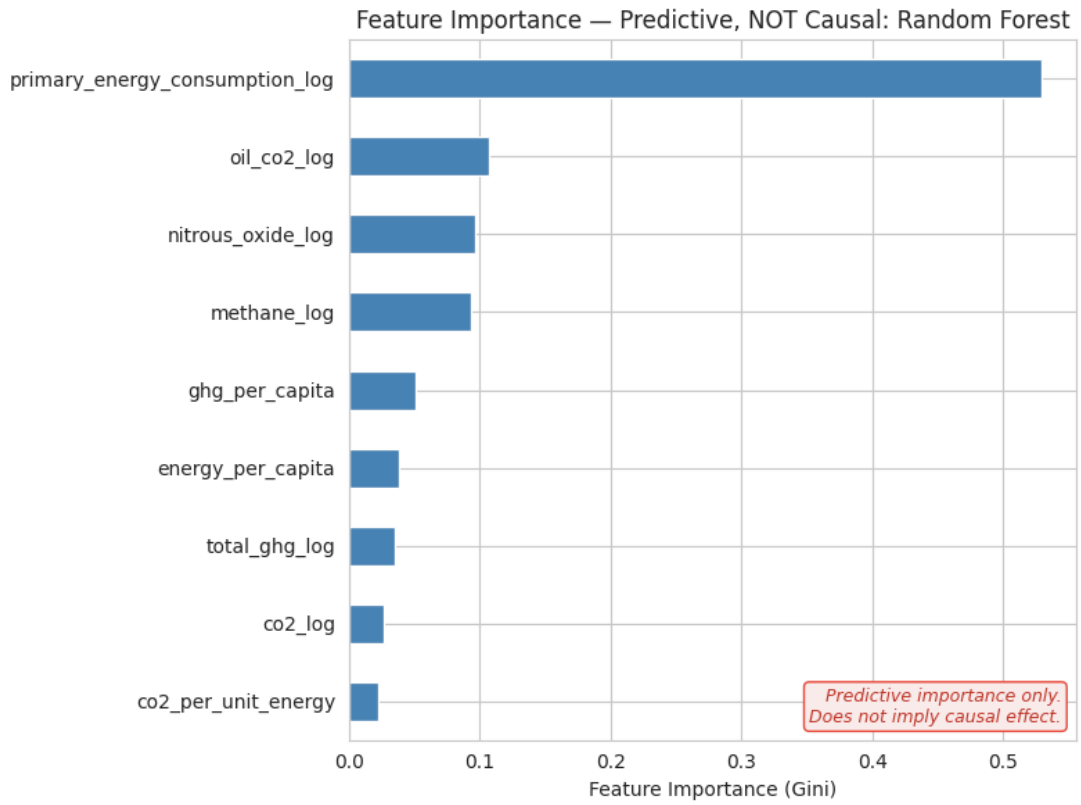

Missing Data Heatmap: Final Cleaning
country
year
iso_code
population
gdp
co2
co2_per_unit_energy
coal_co2
consumption_co2
energy_per_capita
ghg_excluding_lucf_per_capita
ghg_per_capita
land_use_change_co2
methane
nitrous_oxide
oil_co2
primary_energy_consumption
share_global_co2
total_ghg
total_ghg_excluding_lucf
trade_co2
1.0
0.8
0.6
0.4
0.2
0.0

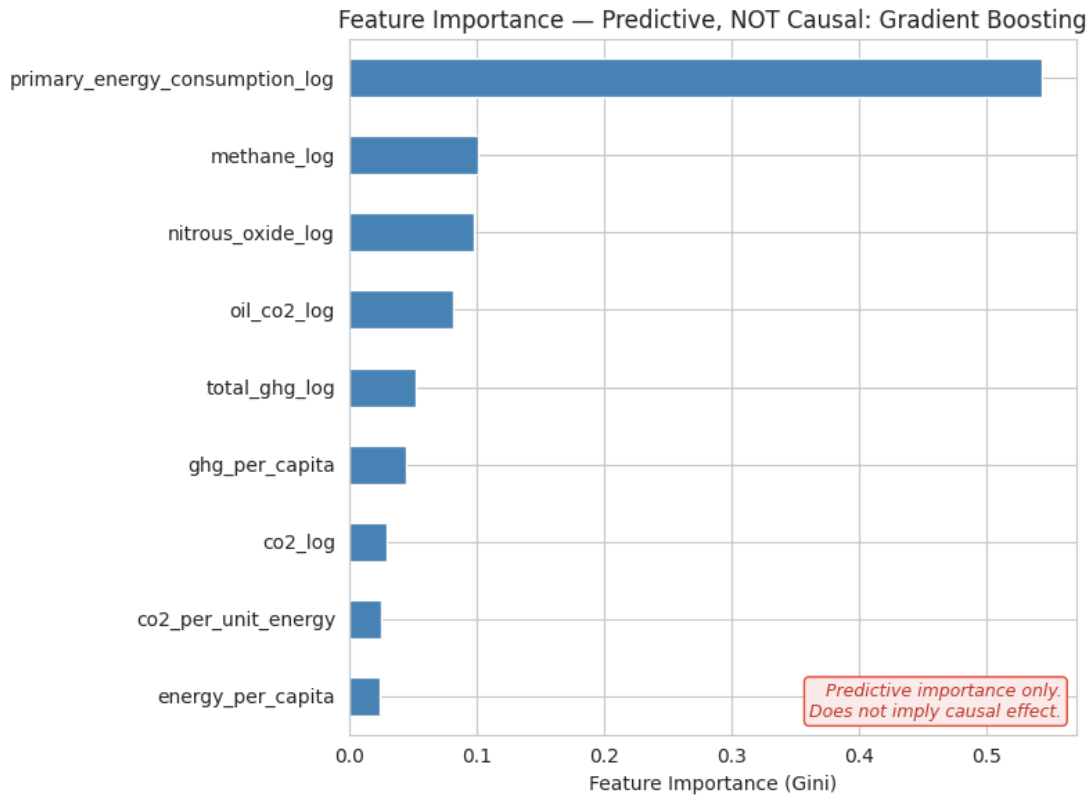

Distribution of Log CO2
Count
co2


Distribution of Log Energy Use per capita
Count
energy_per_capita


Distribution of Log GDP
Count
gdp


Correlation Matrix
gdp
co2
co2_per_unit_energy
energy_per_capita
ghg_excluding_lucf_per_capita
ghg_per_capita
land_use_change_co2
methane
nitrous_oxide
oil_co2
primary_energy_consumption
share_global_co2
total_ghg
total_ghg_excluding_lucf


Feature Importance — Predictive, NOT Causal: Random Forest
primary_energy_consumption_log
oil_co2_log
nitrous_oxide_log
methane_log
ghg_per_capita
energy_per_capita
total_ghg_log
co2_log
co2_per_unit_energy
Predictive importance only.
Does not imply causal effect.
0.0
0.1
0.2
0.3
0.4
0.5
Feature Importance (Gini)


Feature Importance — Predictive, NOT Causal: Gradient Boosting
primary_energy_consumption_log
methane_log
nitrous_oxide_log
oil_co2_log
total_ghg_log
ghg_per_capita
co2_log
co2_per_unit_energy
energy_per_capita
Predictive importance only.
Does not imply causal effect.
0.0
0.1
0.2
0.3
0.4
0.5
Feature Importance (Gini)